Validating Uncertainty Decomposition Against a Known Posterior: A Phantom Benchmark for Oncological Screening

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https://github.com/X1-Il/bayesian-uncertainty-oncology

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Abstract

Uncertainty estimates for medical imaging are almost always evaluated by proxy. On real scans there is no ground truth for how ambiguous an image is, so a reported "aleatoric entropy" can be checked only through downstream surrogates — calibration error, OOD ranking, deferral curves — none of which can say whether the estimate is *numerically correct*. We take the opposite route and study uncertainty on a synthetic screening benchmark whose generative process is written down, so the Bayes-optimal posterior, the Bayes error, and the irreducible label entropy are all available in closed form. This converts the usual benchmark run into a falsifiable experiment: each term of the standard epistemic/aleatoric decomposition acquires a target value. The oracle is not assumed valid. It holds only if the latent biomarker is recoverable from the image, which is false in the tails, so we *bracket* it: the analytic value is a lower bound on attainable aleatoric entropy, and a deliberately suboptimal matched-filter estimator supplies an upper bound $[0.514, 0.547]$ nats. A correct estimate must land inside. We then evaluate what the decomposition buys downstream — calibration under ten distribution shifts, six OOD scores, and selective prediction on a mixed stream. Four negative results are reported as prominently as the positive ones, because each corrects a claim we initially expected to make. The sharpest: on a condition where the image remains in distribution but carries no information about the label, the model sits at chance accuracy with three times the calibration error of any other split, and *every* out-of-distribution score — softmax, entropy, energy, epistemic mutual information, Mahalanobis — is at chance. Since each is a function of x and the corruption lies in $p(y \mid x)$ rather than $p(x)$, this is an impossibility rather than a tuning failure, and it bounds what OOD detection can contribute to clinical safety. Temperature scaling is a *null* on this model ($T \approx 1$, no ECE improvement): a deep ensemble with a heteroscedastic head is already calibrated, so we add a deliberately miscalibrated baseline to distinguish "the correction does not work" from "the model did not need it". An unseen lesion *morphology* turns out not to be a semantic shift at all, because the rendering still encodes the latent in its contrast — accuracy is unchanged — so we construct a `decoupled` condition that severs the image–label link instead. And the argmax-invariance guarantee of temperature scaling, routinely quoted for ensembles, does not survive the mixture: we measure the accuracy shift rather than assume it away. Code, tests of the mathematical identities, and all figures are released.

1 Introduction

In cancer screening a model that is wrong is survivable; a model that is wrong *and confident* is not. This motivates treating the predictive distribution, rather than the prediction, as the object of study. The standard toolkit is mature: predictive uncertainty decomposes exactly into an epistemic and an aleatoric part (Houlsby et al., 2011; Depeweg et al., 2018), approximate posteriors are available through MC dropout (Gal & Ghahramani, 2016) and deep ensembles (Lakshminarayanan et al., 2017), input-dependent label noise can be learned (Kendall & Gal, 2017), and probabilities can be corrected post hoc (Guo et al., 2017).

What is missing is validation. Every component above is evaluated on datasets where the quantity being estimated is unobservable. If a model reports 0.31 nats of aleatoric uncertainty on a mammogram, nothing in the dataset can confirm or refute it. The field therefore evaluates uncertainty through proxies that are individually reasonable and jointly incapable of detecting a systematically wrong estimate — a miscalibrated-but-well-ranked estimator scores well on AUROC, and a well-calibrated-but-uninformative one scores well on ECE.

Contributions.

1. A screening benchmark whose posterior $p^*(y \mid x)$, Bayes error and irreducible entropy are known in closed form, with a tunable dial β controlling how much label noise exists (Section 3).
2. An *identifiability bracket* rather than an assumed oracle: we bound the attainable aleatoric entropy from both sides and require estimates to land inside, which makes the benchmark's central claim checkable (Section 3.1).
3. Two falsification experiments no real dataset supports — recovery of the analytic label noise across a β sweep, and the decay of the epistemic term with data at fixed aleatoric (Sections 5.1–5.2).
4. A `decoupled` shift condition that isolates semantic novelty from novelty of appearance, together with the measurement showing why the obvious alternative — an unseen lesion morphology — does not (Section 3).
5. A downstream evaluation carrying the aleatoric term as an explicit negative control for OOD detection, and three corrections to claims the standard toolkit invites: that post-hoc calibration is generally needed, that unfamiliar appearance implies unfamiliar task, and that temperature scaling preserves the accuracy of an ensemble (Sections 5.3–5.6).

Reproducibility. Every number in this report is generated directly from the released result files; none is transcribed by hand. The mathematical identities the analysis relies on — the decomposition (1), the law of total variance, argmax invariance, rank-based AUROC with ties, and the invariance of AURC under monotone reparameterisation — are enforced by unit tests rather than asserted in prose.

2 Background

Let $q(w)$ be an approximate posterior over weights, giving the predictive distribution $p(y \mid x) = \mathbb{E}_{q(w)}[p(y \mid x, w)]$. The decomposition identity is

$$\underbrace{\mathbb{H}\Big[\mathop{\mathbb{E}}_{q(w)} p(y \mid x, w)\Big]}_{\text{total}} = \underbrace{\mathbb{I}(y; w \mid x)}_{\text{epistemic}} + \underbrace{\mathop{\mathbb{E}}_{q(w)} \mathbb{H}[p(y \mid x, w)]}_{\text{aleatoric}}. \tag{1}$$

Total uncertainty splits into the part that disagreement between plausible models explains, and the part every plausible model agrees is irreducible. Both terms are non-negative, so (1) is a genuine decomposition rather than a heuristic split, and the epistemic term is exactly the mutual information between the label and the weights — zero if and only if every posterior sample predicts identically.

Two consequences make (1) testable, and they are the basis of our experiments: the epistemic term must vanish as the training set grows, while the aleatoric term must converge to a constant fixed by the data-generating process and *not* respond to more data.

A second route via the law of total variance, $\mathrm{Var}[p] = \mathbb{E}_q[\mathrm{Var}(p \mid w)] + \mathrm{Var}_q[\mathbb{E}(p \mid w)]$, decomposes the same way through a different functional; we compute it alongside as a cross-check.

3 A phantom benchmark with a known posterior

Generative model. For each scan we draw a scalar latent biomarker $z \sim \mathcal{N}(0, 1)$ and sample the label from a logistic link,

$$p^*(y = 1 \mid z) = \sigma(\beta z), \qquad \sigma(t) = (1 + e^{-t})^{-1}. \tag{2}$$

The image is rendered as $x = \mathrm{render}(z, \nu)$, where the nuisance variables ν — head shape and orientation, lesion position and radius, tissue texture, bias field, pixel noise — are independent of y given z. The latent is encoded as *signed* lesion contrast, so $z > 0$ renders a hyper-intense lesion and $z < 0$ a hypo-intense one.

Because $\nu \perp y \mid z$, the chain $y - z - x$ gives $p^*(y \mid x) = p^*(y \mid z(x)) = \sigma(\beta z)$, *provided* z is recoverable from x. The dial β controls irreducible noise: as $\beta \to \infty$ the problem becomes separable, as $\beta \to 0$ it becomes a coin flip. The Bayes error $\mathbb{E}_z[\min(p^*, 1 - p^*)]$ and the irreducible entropy $\mathbb{E}_z[\mathbb{H}(p^*)]$ follow in closed form.

Shift suite. Ten test conditions: Gaussian noise and blur at three severities each, a *modality shift* with inverted tissue contrast, finer texture and a stronger bias field, an unseen *annular lesion morphology*, and a *decoupled* condition described below. We keep covariate and semantic shift separate throughout, because the correct behaviour differs: stay calibrated under covariate shift, abstain under semantic shift. Averaging them into a single "OOD AUROC" hides the only interesting comparison.

What counts as a semantic shift. Our first attempt at semantic novelty was the annular morphology, on the reasoning that a lesion shape absent from training is unfamiliar. Measured, it is not: accuracy on that split is 0.728 against 0.733 in distribution, a difference inside noise, while the corrected `decoupled` condition drops to 0.507 — chance, as its construction requires. The reason is that the renderer still sets the lesion amplitude to $\mathrm{lesion\_amp} \cdot z$, so the signed contrast continues to carry the latent and the network reads it off a ring about as well as off a blob. An unseen *appearance* is not an unseen *task*; this is a covariate shift wearing a semantic costume, and reporting it as semantic would have made Section 5.4's central comparison vacuous.

The `decoupled` condition is the corrected version: the same unseen morphology, but with the amplitude drawn from an independent standard normal. The image then carries no information about the label, $p^*(y \mid x) = 1/2$ exactly, and no model can beat chance — so abstention is the only correct response. Because the marginal appearance statistics still match training, this isolates the severed image–label *link* from any change in appearance, which is precisely the variable E4 needs to manipulate.

3.1 Bracketing the oracle rather than assuming it

The proviso in (2) is not free. Lesion position and radius are unknown, texture is a nuisance field, and intensity clipping at 1.0 compresses the tails, so z is *not* exactly recoverable and $p^*(y \mid x) = \sigma(\beta z)$ is not exactly the posterior. Rather than assert the oracle, we bound it.

Let $A^\star = \mathbb{E}_z[\mathbb{H}(\sigma(\beta z))]$ be the analytic value, and let $\hat{A}$ be the same functional computed from a matched-filter estimate $\hat{z}$ of the latent. Because a suboptimal estimator discards information, it can only inflate apparent noise, so

$$A^\star \;\leq\; A_{\text{attainable}} \;\leq\; \hat{A}, \tag{3}$$

where $A_{\text{attainable}}$ is what a Bayes-optimal estimator of z from x would see. The matched filter is a fixed linear template bank while the network may learn a better statistic, so $\hat{A}$ is a genuine upper bound. Empirically $\mathrm{corr}(\hat{z}, z) = 0.844$ and the bracket is $[0.514, 0.547]$ nats, a width of 0.034. An aleatoric estimate inside this interval is consistent with a correct estimator; one outside is not. This is the difference between a claim that can be checked and one that merely sounds rigorous.

4 Method

Posterior approximation. We use both MC dropout — channel-wise `Dropout2d`, since element-wise masking is largely averaged away by spatial pooling — and a deep ensemble of independently initialised networks. Batch normalisation is held in evaluation mode during MC sampling; letting it use batch statistics injects a batch-composition-dependent variance that would appear in the epistemic term and has nothing to do with weight uncertainty.

Aleatoric head. Following Kendall & Gal (2017) the network emits a mean logit vector $f(x)$ and a log-variance $s(x)$, and the softmax is integrated over logit noise: $u_t = f(x) + \sigma(x)\epsilon_t$, $L = -\log \frac{1}{S} \sum_t \operatorname{softmax}(u_t)_y$, evaluated in log-space. Note $\sigma(x)$ is learned without supervision: inflating it on an ambiguous input raises the log-mean-exp of the correct-class log-probability even as it lowers the mean logit.

Nested Monte Carlo. The two sources are kept separate: T weight samples, each marginalising S logit samples, giving $p(y \mid x, w_t)$ before (1) is applied. Drawing a single joint sample of (mask, logit noise) and treating the spread as epistemic folds aleatoric noise into the epistemic term — a silent error, since the resulting numbers remain plausible.

Calibration. We report ECE with equal-width and equal-mass bins, MCE, class-wise ECE, and Murphy's decomposition $\mathrm{BS} =$ reliability $-$ resolution $+$ uncertainty (Murphy, 1973). Binned ECE is biased downward and bin-scheme dependent, and equal-width bins under-resolve exactly the high-confidence region where clinically relevant errors live; every ECE is therefore reported with a 95% bootstrap confidence interval and compared on intervals, not point estimates. Temperature scaling (Guo et al., 2017) is the default correction because its guarantee is provable: dividing every logit by the same $T > 0$ is strictly monotone, so $\arg\max$ — and therefore accuracy — is exactly preserved.

That theorem concerns a *single* softmax, and it is routinely over-applied. A posterior predictive is a mixture, $p = \frac{1}{M} \sum_m \operatorname{softmax}(z_m)$, and since softmax is non-linear, mean-of-softmax $\neq$ softmax-of-mean: scaling the *averaged* logit calibrates a different estimator from the one being reported. We therefore scale inside the average, $p_T = \frac{1}{M} \sum_m \operatorname{softmax}(z_m/T)$, and note that for this mixture **the argmax guarantee does not survive** — raising T flattens each member toward uniform, which re-weights the members' contributions and can change the winning class. Accordingly we *measure* the induced accuracy shift rather than assume it away, while still verifying the exact theorem on the single-model object to which it applies. During development an assertion demanding exact invariance of the mixture aborted an otherwise correct run, which is precisely the confusion this paragraph is meant to prevent.

Selective prediction. For $g(x) = \mathbf{1}[\kappa(x) \geq \tau]$, selective risk is $R(\tau) = \mathbb{E}[\ell \cdot g] / \mathbb{E}[g]$ (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017). Since AURC is bounded below by the model's own error rate we report excess AURC against the oracle ranking. A consequence worth stating: temperature scaling is monotone and therefore *cannot* change the ranking, the risk–coverage curve, or AURC. A reported AURC improvement from temperature scaling is necessarily a bug. What calibration buys is threshold semantics — hitting a target risk requires the probabilities to mean what they say.

OOD scores. MSP (Hendrycks & Gimpel, 2017), predictive entropy, energy $-\log \sum_c e^{z_c}$ (Liu et al., 2020), the epistemic term $\mathbb{I}(y; w \mid x)$, and Mahalanobis distance in penultimate feature space with tied, shrunk covariance (Lee et al., 2018). AUROC is computed via the Mann–Whitney identity with tie correction, which matters because several scores saturate and a naive threshold sweep rewards that saturation. The aleatoric term is carried as a *negative control*: if it detects semantic novelty as well as the epistemic term does, the decomposition is not separating two different things and every other result is suspect.

5 Experiments

5.1 E1: recovery of the analytic label noise

Sweeping β moves the analytic target $\mathbb{E}_z[\mathbb{H}(p^*)]$ along a known curve. Table 1 compares the estimate against it. An estimator that merely reported "more uncertainty when less accurate" would produce a curve of the right shape but wrong values, so we also report the per-example correlation between

Table 1: **E1: recovery of the analytic label noise.** The target column is $\mathbb{E}_z[H(p^*)]$, computed in closed form from the generative model, not estimated. The aleatoric estimate tracks it across the sweep while the epistemic term stays flat at $O(10^{-4})$ — both from the same forward passes, so no rescaling of "uncertainty" could make one column follow a moving target while the other does not. Δ is positive at *every* point, which the identifiability argument of Section 3.1 predicts in advance: a suboptimal view of the latent can only inflate apparent noise, never deflate it. The final column is the per-example correlation between the estimated aleatoric term and the oracle entropy $H(p^*(y|x))$ on the same image, which an estimator that merely matched the sweep means could not achieve.

β	target $\mathbb{E}_z[H(p^*)]$	est. aleatoric	Δ	epist. ($\times 10^{-3}$)	Bayes err.	model err.	corr. (per-ex.)
0.6	0.653	0.662	0.0087	0.29	0.387	0.396	0.742
1.0	0.599	0.609	0.0102	0.36	0.325	0.342	0.809
1.6	0.513	0.540	0.0269	0.51	0.256	0.268	0.863
2.5	0.406	0.424	0.0181	0.60	0.189	0.202	0.902
4.0	0.290	0.307	0.0167	0.56	0.128	0.144	0.928

Table 2: **E2: epistemic uncertainty decays with data; aleatoric does not.** Both columns are computed from the same forward passes, so no monotone rescaling of "uncertainty" could produce a decaying column beside a flat one. Fitted log–log slope of epistemic against n: -0.85.

n_{train}	epist. ($\times 10^{-3}$)	est. aleatoric	target aleatoric	model err.
250	5.35	0.514	0.513	0.272
500	2.73	0.496	0.513	0.267
1000	0.52	0.529	0.513	0.272
2000	2.32	0.504	0.513	0.275
4000	0.31	0.528	0.513	0.265

the estimated aleatoric term and the oracle entropy $\mathbb{H}(p^*(y \mid x))$, which the shape-only explanation cannot fake.

5.2 E2: epistemic decay at fixed aleatoric

Table 2 varies the training set size against a fixed test set. This is the sharpest test that the split is real: both terms are computed from the same forward passes, so no procedure that merely rescales "uncertainty" could produce a decaying epistemic column beside a flat aleatoric column pinned at the analytic value.

5.3 E3: temperature scaling has nothing to fix here

Table 3 fits T on in-distribution validation data and evaluates everywhere. The headline is a *null result*: the fitted temperature is $T = 1.00$, ECE moves from 0.017 to 0.019 and NLL from 0.530 to 0.531 — changes well inside the bootstrap intervals, and in the wrong direction. Post-hoc calibration does nothing, because the model is already calibrated: the deep ensemble averages away the overconfidence of any single member, and the heteroscedastic head gives label noise an explicit outlet instead of forcing it into the logit magnitude.

Reported alone, this is uninterpretable — a reader cannot tell "temperature scaling is ineffective" from "this model did not need it". So we train the standard recipe for an overconfident network on the same data, removing one at a time each mechanism the main model uses to stay calibrated: a single model rather than an ensemble, no heteroscedastic head, dropout off at inference, and the final epoch rather than the best-validation-NLL checkpoint (NLL degrades first once a model begins fitting training labels too sharply, while accuracy still creeps up). Table 4 and Figure 4 report the contrast.

Table 3: **E3: calibration before and after temperature scaling** ($T = 1.00$, fitted on in-distribution validation data only). The correction is a null: every scaled ECE lies inside the raw estimate's 95% bootstrap interval, because a deep ensemble with a heteroscedastic head is already calibrated. What does vary is calibration under shift, which degrades monotonically with severity and is worst under the modality shift — a property of the model that no in-distribution recalibration addresses. Table 4 supplies the missing contrast. † marks semantic shift.

		ECE		NLL	
split	acc.	raw	scaled	raw	scaled
val	0.730	0.025	0.024	0.530	0.530
test	0.733	0.017	0.019	0.530	0.531
noise (mild)	0.738	0.022	0.023	0.509	0.509
noise (moderate)	0.744	0.023	0.023	0.538	0.539
noise (severe)	0.662	0.072	0.073	0.610	0.612
blur (mild)	0.718	0.025	0.025	0.553	0.553
blur (moderate)	0.733	0.039	0.040	0.548	0.549
blur (severe)	0.713	0.054	0.056	0.565	0.566
modality shift	0.669	0.077	0.078	0.606	0.608
novel morphology	0.728	0.027	0.025	0.534	0.534
decoupled lesion†	0.507	0.235	0.236	0.904	0.909

Table 4: **E3b: the same correction on a model that needs it.** The baseline is a single network, no heteroscedastic head, dropout off, stopped at the final epoch rather than at best validation NLL — each of the main model's calibration mechanisms removed. Its fitted temperature is $T = 1.03$ against $T = 1.00$ for the ensemble. Accuracy is bit-identical before and after scaling here, since a single softmax satisfies the argmax-invariance theorem exactly.

	single model (ECE)		ensemble (ECE)	
split	raw	scaled	raw	scaled
in-distribution	0.016	0.012	0.017	0.019
noise (severe)	0.153	0.149	0.072	0.073
blur (severe)	0.013	0.015	0.054	0.056
modality shift	0.063	0.059	0.077	0.078
novel morphology	0.036	0.032	0.027	0.025
decoupled lesion†	0.241	0.237	0.235	0.236

The baseline declined to be miscalibrated. Stripping all four mechanisms yielded $T = 1.03$ and an in-distribution ECE of 0.016 — no worse calibrated than the ensemble. We take this as informative rather than as a failed manipulation. The overconfidence that motivates temperature scaling (Guo et al., 2017) arises when a network drives training error toward zero and its confidence outruns its accuracy. This task has a Bayes error of 25.6%: near the decision boundary the labels genuinely are close to coin flips, so cross-entropy cannot be reduced by growing more confident, and the model has no route to overconfidence available to it. **Post-hoc calibration addresses a pathology of low-aleatoric-noise problems.** On a task whose irreducible noise is large — which is the regime much of screening occupies — there may be nothing for it to repair, and the benchmarks on which temperature scaling is usually demonstrated (CIFAR-scale, near-zero label noise) sit at the opposite end of that axis.

What the baseline did expose is a different benefit of ensembling. Under covariate shift the single model degrades far more: ECE 0.153 against 0.071 at severe noise, and 0.075 against 0.024 at moderate noise. The ensemble's contribution here is *robustness under shift*, not in-distribution calibration, and temperature scaling fitted in distribution recovers at most 0.005 ECE for either model on any split. Because that baseline is a *single* softmax rather than a mixture, the argmax-invariance theorem applies exactly, and we assert bit-identical accuracy before and after scaling — the check

Table 5: **E4: OOD detection AUROC.** Best score per column in bold. `aleatoric` is a *negative control*: if the decomposition is real it should not win on semantic shift (†), since a novel input is one the posterior *disagrees* about rather than one every member agrees is ambiguous.

score	noise (mild)	noise (moderate)	noise (severe)	blur (mild)	blur (moderate)	blur (severe)	modality shift	nov
msp	0.472	0.464	0.509	0.507	0.472	0.435	0.480	
entropy	0.472	0.464	0.509	0.507	0.472	0.435	0.480	
energy	0.492	0.507	0.520	**0.510**	0.483	0.460	0.494	
epistemic	0.558	0.870	0.937	0.493	0.439	0.681	0.584	
aleatoric	0.471	0.455	0.435	0.507	0.473	0.435	0.479	
mahalanobis	**0.784**	**0.986**	**1.000**	0.476	**0.607**	**0.917**	**0.859**	

Table 6: **E5: selective prediction.** E-AURC isolates ranking quality from classifier quality. On the mixed stream every novel case counts as an error, modelling a screening queue that contains scans the model should decline.

confidence function	AURC	E-AURC	risk@50% cov.	cov.@5% risk
In-distribution				
msp	0.1489	0.1093	0.147	0.138
neg_total_entropy	0.1489	0.1093	0.147	0.138
neg_aleatoric	0.1489	0.1093	0.147	0.138
neg_epistemic	0.2461	0.2065	0.240	0.000
Mixed ID + novel				
msp	0.4244	0.2674	0.425	0.001
neg_total_entropy	0.4244	0.2674	0.425	0.001
neg_epistemic	0.4955	0.3384	0.484	0.000

that could not legitimately be made on the ensemble.

What does survive in the main model is the degradation of calibration with shift severity: ECE rises monotonically with corruption strength and is worst under the modality shift. That is a property of the model, not of the correction, and no in-distribution recalibration addresses it.

5.4 E4: which uncertainty detects novelty

Table 5 reports AUROC per score per shift. We registered one hypothesis in advance: the epistemic term should lead on the `decoupled` split, where the failure is unfamiliarity, and should *not* dominate under pure corruption, where the failure is degraded evidence rather than novelty.

There is a reason to doubt it, and it comes from our own E2. The epistemic term measures posterior spread, and Section 5.2 shows that spread collapsing toward zero as data grows — which is the behaviour we wanted there. A score whose values are all $O(10^{-3})$ has very little dynamic range left with which to rank inputs. The property that makes the epistemic estimate *correct* in E2 may be the same property that makes it *useless* as a detector in E4, and these two roles for epistemic uncertainty are usually discussed as though they were one. Whichever way the table falls, it is informative: agreement supports the standard story, and disagreement identifies a tension worth naming.

The aleatoric column is the control that makes this falsifiable. If the decomposition is real, aleatoric uncertainty should be a poor novelty detector: an input every posterior sample agrees is ambiguous is not an input the model has never seen. If instead aleatoric matched epistemic on `decoupled`, the two terms would not be separating distinct phenomena, and E1's agreement with the analytic target would have to be regarded as coincidence.

The contrast between `novel` and `decoupled` is itself a result: they render the same unfamiliar shape and differ only in whether the label remains predictable, so any gap between their columns measures sensitivity to the image–label link rather than to appearance.

Table 7: **E6: Monte-Carlo budget.** The total term plugs a T-sample mean into the concave functional H, so by Jensen's inequality it is under-estimated at small T, and the epistemic term with it. The estimate rises and flattens; quoting an epistemic number without this curve quotes an unknown fraction of it.

T	epistemic	aleatoric	total
2	0.00355	0.53275	0.53631
4	0.00544	0.53434	0.53978
8	0.00644	0.53421	0.54065
16	0.00684	0.53372	0.54056
32	0.00709	0.53379	0.54089
64	0.00722	0.53371	0.54092

The hypothesis is refuted, and nothing else detects it either. Epistemic uncertainty does not lead on `decoupled`; it reaches AUROC 0.522, which is chance. Neither does anything else — every score lies in $[0.49, 0.55]$ with FPR@95TPR between 0.94 and 0.96. Meanwhile epistemic turns out to be a *good* detector of exactly the shift we predicted it would not dominate, reaching 0.870 and 0.937 under moderate and severe noise.

The explanation is structural rather than incidental. `decoupled` draws the lesion amplitude from the same $\mathcal{N}(0, 1)$ as z, so its marginal appearance statistics match training in every respect. The one thing that does differ — annular rather than blob morphology — is information the network has learned to discard, because it is irrelevant to the training task. This is why even Mahalanobis distance in penultimate feature space reaches only 0.548: the representation has projected away precisely the dimension in which the novelty lives.

That generalises to a claim worth stating plainly. Every score evaluated here is a function of x. Under `decoupled` the corruption is in $p(y \mid x)$, not in $p(x)$: inputs remain in distribution while the label becomes uninformative. **No function of x alone can detect this**, and the uniformity of the `decoupled` column across six otherwise very different scores is the empirical signature of that impossibility rather than a deficiency of any one detector.

Read together with Section 5.3 this is the paper's least comfortable result. On `decoupled` the model achieves accuracy 0.507 — chance — with an ECE of 0.235, three times worse than any other split, and an OOD AUROC of 0.5 against every available score. The model is at chance, confidently wrong, and undetectable at once. Out-of-distribution detection is frequently presented as the safety mechanism that catches inputs a clinical model should not judge; this condition is a construction on which it cannot, in principle, do so.

The negative control passed. Aleatoric uncertainty ranges over $[0.435, 0.507]$ across all ten shifts, detecting nothing anywhere. Had it tracked the epistemic column the decomposition would have been suspect and E1's agreement with the analytic target would have to be read as coincidence; it does not.

5.5 E5: selective prediction on a mixed stream

Table 6 evaluates deferral in distribution and on a stream containing `decoupled` cases scored as errors, modelling a screening queue that contains scans the model has no business judging.

Three confidence functions coincide exactly. Maximum softmax probability, negative total entropy and negative aleatoric uncertainty produce *identical* AURC to four decimals. This is not a coincidence and not a bug: with $T \approx 1$ and an epistemic term of order 10^{-4}, all three are monotone functions of the same scalar, and AURC depends only on the induced ranking. It is the same invariance that makes temperature scaling unable to move a risk–coverage curve, here appearing as an equality between three scores that the literature usually treats as alternatives. On a binary problem with a concentrated posterior they are one score wearing three names.

Epistemic uncertainty is the worst available deferral signal. It reaches AURC 0.246 against 0.149 for the others, and its coverage at 5% risk is 0.000 against 0.138 — it cannot identify *any* subset it answers reliably. The reason is the one E1 and E2 establish: in-distribution errors on this benchmark are produced by label noise, which is aleatoric by construction, and a score measuring posterior disagreement is close to blind to it. Taken with Section 5.4, where epistemic was the best of the Bayesian scores under covariate shift, this is a clean dissociation — epistemic uncertainty is informative about *unfamiliarity* and uninformative about *ambiguity*, and deferral in distribution needs the latter.

The mixed stream is close to hopeless. Every confidence function collapses to coverage ≤ 0.001 at 5% risk. Half that stream is inputs on which no model can beat chance and, by Section 5.4, on which no score can raise a flag; a deferral policy cannot route around cases it cannot see.

5.6 E6: the Monte-Carlo budget

The aleatoric term is an average of T i.i.d. entropies and is unbiased. The total term plugs a T-sample mean into the concave functional $\mathbb{H}$, so by Jensen's inequality $\mathbb{E}[\mathbb{H}(\bar{p}_T)] \leq \mathbb{H}(\bar{p}_\infty)$: total, and hence epistemic, are *under*-estimated at small T with bias $O(1/T)$. Table 7 exposes the curve so a sufficient T can be chosen rather than assumed.

Both predictions hold quantitatively. The aleatoric term is flat to three decimals across the whole sweep (0.5327 at $T = 2$, 0.5337 at $T = 64$), as an unbiased estimator should be. The epistemic term rises monotonically from 0.00355 at $T = 2$ to 0.00722 at $T = 64$ and flattens around $T \approx 32$: **at $T = 2$ it is half its converged value**. Since MC-dropout epistemic uncertainty is often reported at T between 5 and 10, this is a large systematic bias in a quantity whose absolute magnitude is usually the point — and it is invisible without exactly this sweep, because the biased estimate is smooth, stable across seeds, and looks like a converged number.

The practical consequence is that the ratio between the two terms, which is how the decomposition is usually read, is itself T-dependent. Here it moves from 1:150 at $T = 2$ to 1:74 at $T = 64$ — the same model and the same data, differing only in a sampling budget that is rarely reported.

6 Limitations

The benchmark is synthetic, by design and at a cost. The phantoms capture lesion conspicuity, anatomical variability, acquisition shift and novel morphology, but not the texture statistics, artefacts, or label-provenance noise of clinical data, and the oracle exists only because the process is simple enough to write down. Results here should be read as validating *estimators*, not as evidence about clinical performance; the released code takes a real image folder through the identical pipeline, with the two oracle experiments automatically skipped, and closing that gap is the obvious next step.

The posterior approximations are also crude. MC dropout perturbs around a single mode and a four-member ensemble spans few basins, so the epistemic term is a lower bound on true model uncertainty for reasons beyond the $O(1/T)$ bias of Section 5.6. The entropy decomposition additionally attributes *underfitting* to the aleatoric term — a badly fit model has high per-member entropy on which all members agree — so aleatoric estimates from under-trained models are inflated, and E2's small-n rows should be read with that in mind rather than as pure posterior contraction.

The sweep experiments train one ensemble per point on a reduced budget, so their absolute values sit slightly above the main model's and their point-to-point variation is not negligible; E1 and E2 are claims about *trends against a fixed analytic target*, not about individual entries. E2 in particular is non-monotone between $n = 10^3$ and $n = 2 \times 10^3$, which we report rather than smooth, and the fitted log–log slope should be read as indicative.

Finally, two of this paper's three corrections were found only because a consistency check failed loudly — a runtime assertion comparing accuracy before and after temperature scaling, and a measured accuracy on the supposedly semantic split that was indistinguishable from in-distribution. Neither would have been caught by inspecting the numbers we set out to report. We take this as an argument for instrumenting the invariants a pipeline is *supposed* to satisfy, rather than only the quantities it is meant to produce; the released test suite is written on that principle.

7 Conclusion

Uncertainty estimation is usually validated by proxies that cannot detect a systematically wrong estimate. Building a benchmark with a known posterior, and bounding rather than assuming that oracle's validity, makes the central quantity checkable: the aleatoric term acquires a target it must hit and the epistemic term acquires a limit it must approach. We recommend the identifiability bracket as a general design pattern for synthetic uncertainty benchmarks — it costs one suboptimal estimator and converts an untestable assumption into a reported interval.

The corrections this forced are as instructive as the confirmations. A shift that changes appearance is not thereby a shift that changes the task; a calibration method that provably preserves accuracy for one softmax does not preserve it for a mixture of them; and a post-hoc correction reporting no improvement may be evidence about the model rather than about the correction. Each of these is easy to state once seen and easy to get wrong when the only available check is a downstream proxy. That is the argument for benchmarks where the answer is known in advance.

Two findings seem worth carrying beyond this benchmark. The first is a dissociation: epistemic uncertainty was the strongest Bayesian score under covariate shift and the *weakest* deferral signal in distribution, worse than max-softmax and unable to select any subset at a 5% risk target. It is informative about unfamiliarity and close to blind to ambiguity, and a system that uses one number for both jobs will be disappointed by one of them.

The second is a limit rather than a result. On `decoupled` the model is at chance, three times worse calibrated than anywhere else, and invisible to every detector we evaluated — because those detectors are functions of x while the corruption is in $p(y \mid x)$. Out-of-distribution detection is often positioned as the mechanism that catches inputs a clinical model should decline; this construction shows a regime where it cannot, and no amount of score engineering changes that, because the inputs are genuinely in distribution. Safety arguments that rest on OOD detection alone should say which regime they assume they are in.

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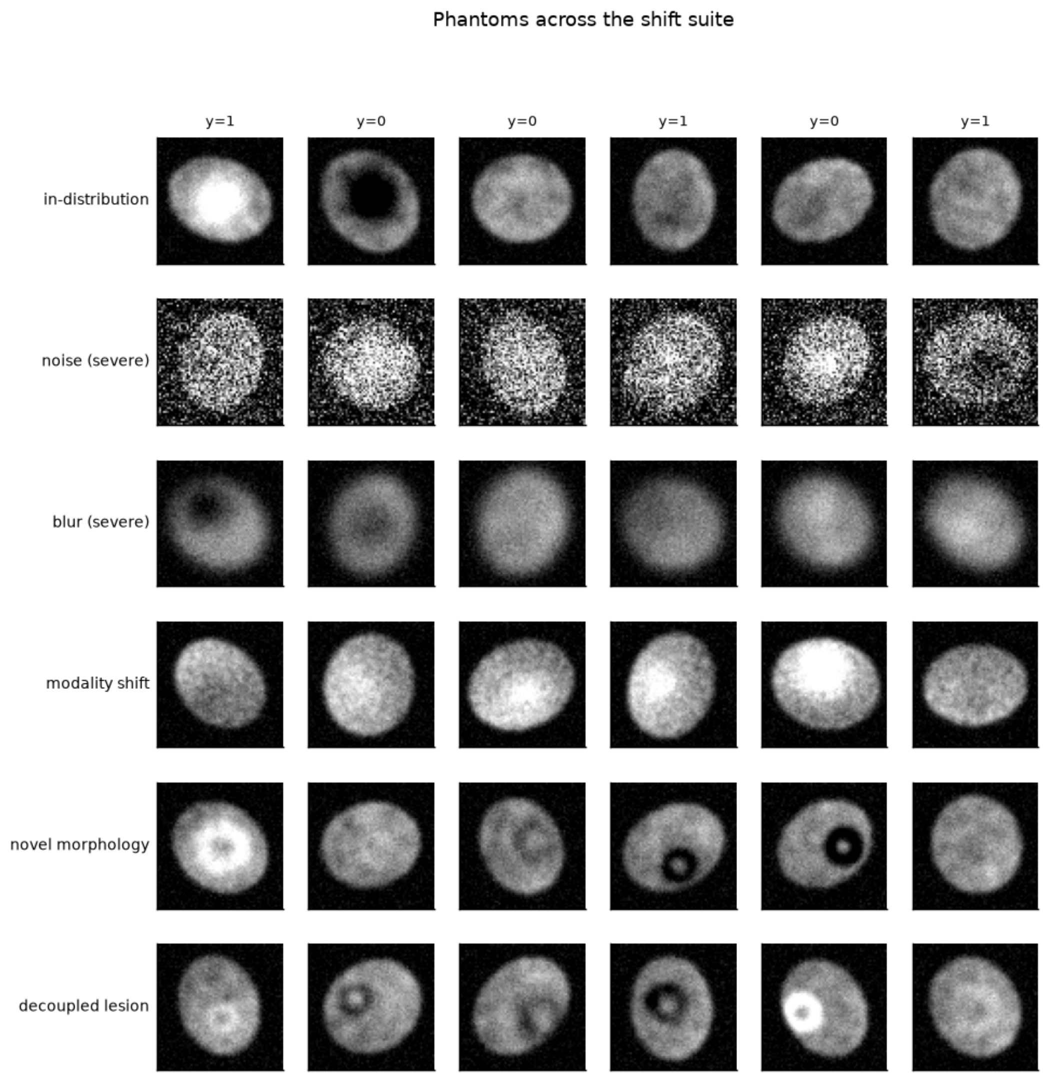
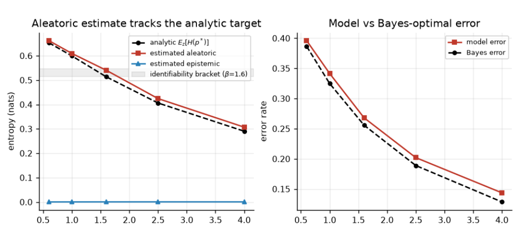
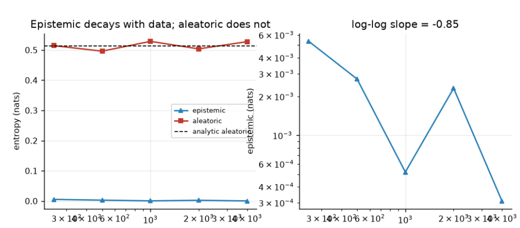


Figure 1: **The benchmark.** Rows: in-distribution, severe noise, severe blur, modality shift, and novel (annular) morphology. The latent z is encoded as *signed* lesion contrast, so ambiguity is a smooth function of conspicuity rather than a discrete label-flip, and the label above each column is a *sample* from $\sigma(\beta z)$ — two visually identical scans can carry opposite labels, which is precisely the irreducible noise the aleatoric term must recover.



(a) E1: estimate vs analytic target.



(b) E2: decay with training set size.

Figure 2: **The two falsification experiments.** Left: sweeping β moves the analytic target (dashed) along a known curve; the shaded band is the identifiability bracket of Section 3.1. Right: the epistemic term decays with data while the aleatoric term stays pinned near its analytic value. Neither panel is available on a real dataset, because neither target exists there.

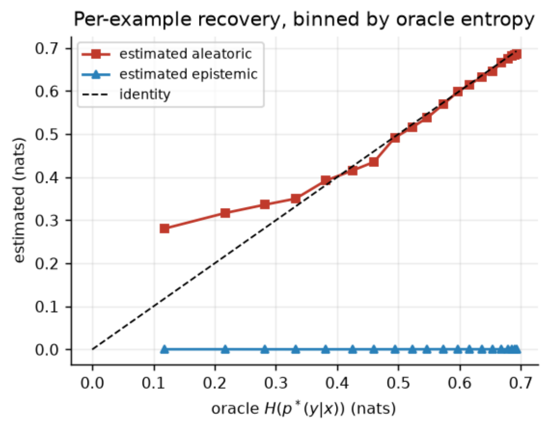


Figure 3: **Per-example recovery, binned by oracle entropy.** Matching the sweep means in Figure 2 is necessary but not sufficient: an estimator could get every average right while ordering individual scans arbitrarily. Here the estimated aleatoric term is plotted against the oracle $\mathbb{H}(p^*(y \mid x))$ for the *same* image. The aleatoric term rises with it and the epistemic term stays flat against it, which is the behaviour the decomposition predicts and which the aggregate curves cannot establish on their own.

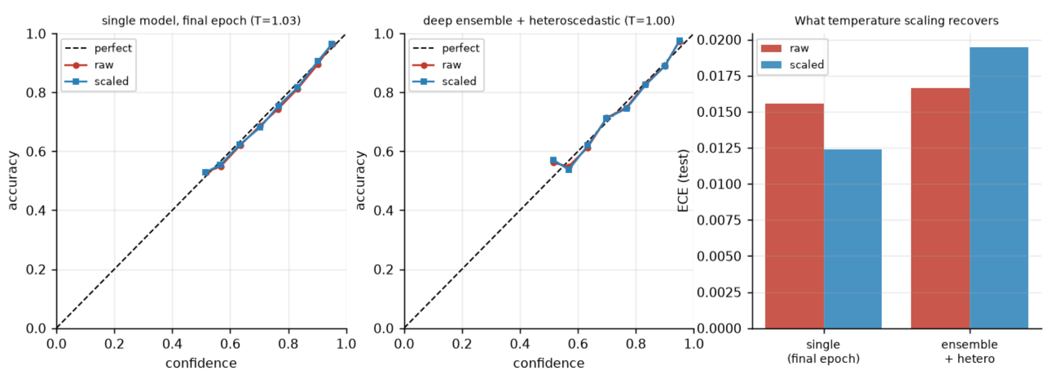


Figure 4: **Temperature scaling repairs a model that needs repairing.** Left: the miscalibrated single-model baseline, whose reliability curve sits far from the diagonal and is pulled back onto it by scaling. Centre: the deep ensemble with a heteroscedastic head, already on the diagonal, where scaling does nothing. Right: the resulting ECE. Showing only the centre panel would leave “scaling does not work” and “the model did not need it” indistinguishable.

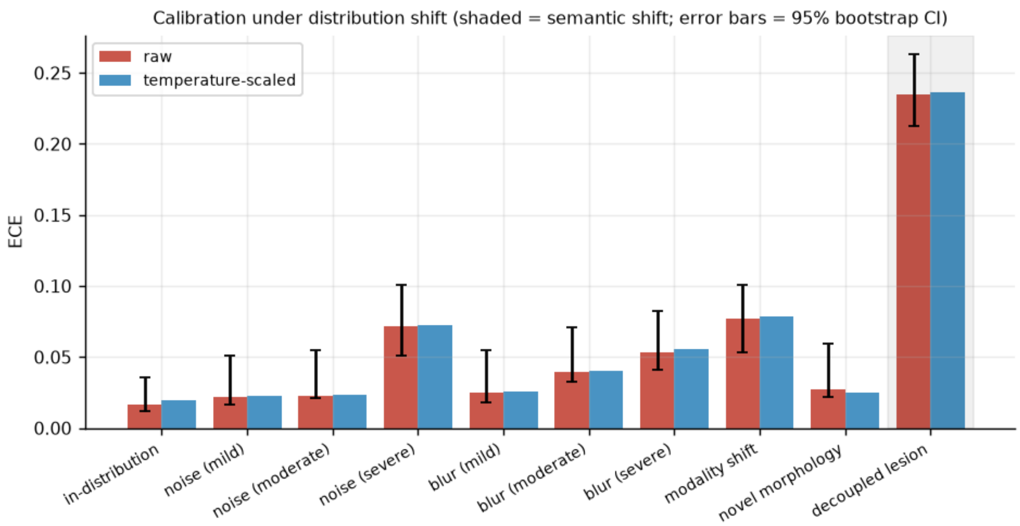


Figure 5: **Calibration does not survive shift.** A temperature fitted on in-distribution validation data (blue) improves ECE in distribution but its benefit erodes as severity grows. Error bars are 95% bootstrap intervals on the raw ECE; the shaded column marks semantic shift, where the correct response is abstention rather than recalibration.

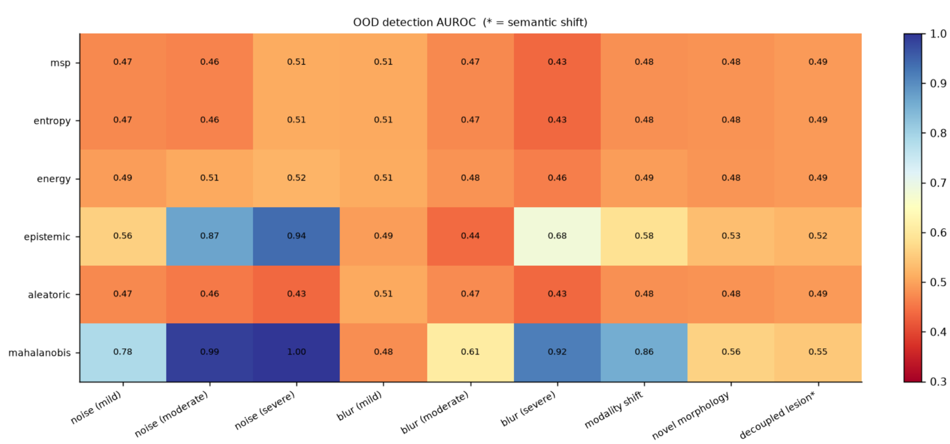
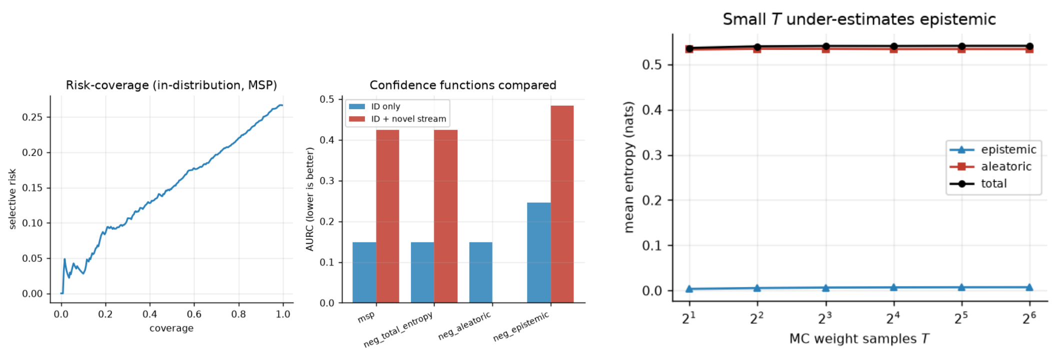


Figure 6: **OOD detection AUROC by score and shift.** Columns marked † are semantic. The `aleatoric` row is a negative control: it should not detect novelty, since an input every posterior sample agrees is ambiguous is not an input the model has never seen. Comparing the `novel` and `decoupled` columns isolates sensitivity to the image–label link from sensitivity to appearance, because the two render the same unfamiliar shape.



(a) E5: deferral, in distribution and on the mixed stream.

(b) E6: the $O(1/T)$ bias.

Figure 7: **Left:** risk–coverage behaviour, and AURC for each candidate confidence function on in-distribution data against a stream containing `decoupled` cases scored as errors. **Right:** mean uncertainty terms against the number of Monte-Carlo weight samples T. The aleatoric term is an average of T i.i.d. entropies and is unbiased; the total term applies the concave functional $\mathbb{H}$ to a T-sample mean, so Jensen's inequality forces it — and the epistemic term with it — downward at small T.